



University
Mohammed VI
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Deliverable #: Lab3 (e.g.: relational algebra)

Data Management Course
UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
Session: Fall 2026

Team Information

Team Name	role
TAHA YASSINE	Q1, Q6, Q10, Q14
SAAD SABER	Q2, Q5, Q8, Q13
HAFFSA SAHNOUNE	Q3,Q7,Q11,Q15
ACHRAF RHANNAM	Q4, Q9, Q12
Repository Link	https://github.com/...

1 queries SOLUTION

1.1 Q1, Q6, Q10, Q14

1. $\pi_{\text{firstName,lastName}}(Person \bowtie_{Person.pid=Reservation.madebypid} Reservation)$
2. $T = FundsPrj \bowtie_{FundsPrj.budgetline=fundslab.budgetline} (\sigma_{FundsLab.labId=2}(FundsLab))$

$$\pi_{pid,codeProject}(Participate) / \pi_{codeProject}(T)$$

3. $E1 = \rho_{E1}(\sigma_{role="PI"}(Participate))$
 $E2 = \rho_{E2}(\sigma_{role="PI"}(Participate))$

$$\pi_{pid}(E1 \bowtie_{E1.pid = E2.pid \text{ and } E1.codeProject \neq E2.codeProject} E2)$$

4. $C1 = \pi_{labId,serialNo}(EquipmentUnit)$
 $C2 = \pi_{restId,MadeByPid}(\sigma_{status="Approved"}(Reservation))$
 $C3 = \pi_{labId,restId,madeByPid}(C2 \bowtie Reserves \bowtie C1)$
 $C4 = \gamma_{labId,madeByPid;COUNT(restId)->n}(C3)$
 $C5 = \gamma_{labId;MAX(n)->m}(C4)$

$$R = \pi_{labId,madeByPid}(C4 \bowtie_{C4.labId=C5.labId \text{ and } C4.n=C5.m} C5)$$

1.2 Q3, Q7, Q11, Q15

1. $C1 = \pi_{labId}(\sigma_{building="College of Computing"}(Laboratory))$
 $C2 = \pi_{labId}(\sigma_{category="CPU Server"}(EquipmentModel) \bowtie EquipmentUnit)$

$$R = C1 \cup C2$$

2. $Counts = \gamma_{pid;COUNT(resId)->n}(Person \bowtie_{Person.pid=Reservation.madeByPid} Reservation)$
 $C1 = \rho_{C1}(Counts)$
 $C2 = \rho_{C2}(Counts)$

$$R = \pi_{C1.pid->p1,C2.pid->p2}(\sigma_{C1.n>C2.n}(C1 \times C2))$$

3. $A = \pi_{resId,approverPid}(\sigma_{approverPid \neq NULL}(Reservation))$
 $B = \pi_{serialNo,labId}(EquipmentUnit)$
 $AL = \pi_{approverPid,labId}(A \bowtie Reserves \bowtie B)$
 $A1 = \rho_{A1}(AL)$
 $A2 = \rho_{A2}(AL)$

$$R = \pi_{A1.approverPid}(A1 \bowtie_{A1.approverPid = A2.approverPid \text{ and } A1.labId \neq A2.labId} A2)$$

4. $R0 = \sigma_{submissionTS \geq "2025-01-01" \text{ and } submissionTS < "2026-01-01"}(Reservation)$
 $R25 = \pi_{resId,madeByPid}(R0)$
 $C = \gamma_{madeByPid;COUNT(resId)->n}(R25)$

$$R = \pi_{madeByPid}(\sigma_{n \geq 3}(C))$$

1.3 Q2, Q5, Q8, Q13

1.

$$\rho_{PR}(\text{Participate}) \quad \rho_A(\text{Attached})$$

$$\pi_{pid} (PR \bowtie_{PR.Role="PT"} A)$$

2.

$$\rho_S(\text{Supervises}) \quad \rho_L(\text{Laboratory})$$

$$\pi_{pid} (S \bowtie_{L.building="College of Computing"} L)$$

3.

$$\rho_{Res}(\text{Reservation})$$

$$\rho_{Res}(\text{Reserves})$$

$$\rho_{EU}(\text{EquipmentUnit})$$

$$T = \pi_{madeByPid,labId} (Res \bowtie_{resId} Res \bowtie_{serialNo} EU)$$

$$T_1(pid_1, lab_1) \quad : \quad T_2(pid_2, lab_2)$$

$$\pi_{pid_1} (\sigma_{(pid_1=pid_2) \wedge (lab_1 \neq lab_2)} (T_1 \times T_2))$$

4. We are going to use the $\mathcal{G}$ sign, which means the aggregation sign:

$$LabCounts = labId \mathcal{G}COUNT(serialNo) \rightarrow unitCount (L \bowtie_{labId} EU)$$

$$AvgUnits(avgCount) = \emptyset \mathcal{G}AVG(unitCount) \rightarrow avgCount(LabCounts)$$

$$\pi_{labId} (\sigma_{unitCount < avgCount} (LabCounts \times AvgUnits))$$

1.4 Q4, Q9, Q12

1.

$$\pi_{labId} \left(\pi_{labId} (\text{EquipmentUnit} \bowtie \sigma_{category='CPU Server'}(\text{EquipmentModel})) \cap \right.$$

$$\left. \pi_{labId} (\text{EquipmentUnit} \bowtie \sigma_{category='GPU Server'}(\text{EquipmentModel})) \right)$$

2.

$$\rho(R_1, \pi_{resId}(\sigma_{submissionTS \geq '2026-09-01' \wedge submissionTS \leq '2026-09-30'}(Reservation)))$$

$$\rho(R_2, \pi_{resId}(Reserves \bowtie EquipmentUnit \bowtie \sigma_{name='AI Movement'}(Laboratory)))$$

$$R_1 \cap R_2$$

3.

$$\pi_{pid}(\text{Person}) - \pi_{madeByPid}(\sigma_{plannedStart \leq '2026-11-06 23:59:59' \wedge plannedEnd \geq '2026-11-06 00:00:00'}(Reservation))$$

Key Challenges and How They Were Addressed

1. **Schema fidelity:** attribute names had to match Lab 2 exactly, so we checked each one and defined prefixes with ρ .
2. **PI information (Q10):** the schema has no PI column, so we selected role = 'PI' through Participates and stated this assumption.
3. **“Every” (Q6):** a plain join gives “at least one”, so we used division ($\div$), or product and difference.
4. **“More than one” (Q10):** basic RA cannot count, so we self-joined the PI rows on the same pid with different projectCode.
5. **“Greatest” per group (Q14):** we projected before counting to keep reservations distinct, grouped by (labId, madeByPid), then joined back on the maximum to recover the person.
6. **Assumptions:** status = 'Approved', ties return all tied persons, and duplicate names in Q1 collapse unless pid is projected.
7. **“Or” and union compatibility (Q3):** each branch of the union projects only labId so both have the same schema.
8. **Counting and zero counts (Q7):** basic RA cannot count, so we used γ . A left outer join with Person keeps persons with no reservation, with $n = 0$.
9. **Comparing two tuples (Q7, Q11):** a selection only tests one tuple, so we used a self cross product with renaming (ρ).
10. **Attribute name clashes (Q11):** Reservation and EquipmentUnit both have status, so we projected before joining.
11. **Dates (Q15):** “made during 2025” uses submissionTS (RE2), with a half-open interval.
12. **Q(2):** Filtered records with Role = "PI" from Participate and joined them with Attached using explicit relation renaming (ρ_{PR} and ρ_A) to project matching person IDs (pid).
13. **Q(5):** Expressed the supervisor relationship for target laboratories by filtering L.building = "College of Computing" and joining Supervises and Laboratory via renamed relations (ρ_S and ρ_L).
14. **Q(8):** Evaluated the multi-row condition (reserving equipment in at least two different laboratories) by joining Reservation, Reserves, and EquipmentUnit, then taking a self-cross-product ($T_1 \times T_2$) with the selection condition $pid_1 = pid_2 \wedge lab_1 \neq lab_2$.
15. **Q(13):** Handled aggregate counts using the $\mathcal{G}$ operator, calculating LabCounts via a Left Outer Join to preserve laboratories with zero equipment, computing AvgUnits across all laboratories, and selecting those where unitCount < avgCount.
16. **Q(4):** Addressed the “at least one of X and at least one of Y” condition by independently filtering EquipmentModel for ‘CPU Server’ and ‘GPU Server’, joining each

with `EquipmentUnit`, projecting the `labId`, and using the intersection operator ($\cap$) to ensure the laboratories contain both.

17. **Q(9):** Divided the query into two independent sets using the renaming operator (ρ): R_1 for reservations submitted in September 2026, and R_2 for reservations involving the 'AI Movement' laboratory. Applied the intersection operator ($\cap$) to return only the `resIds` satisfying both conditions.
18. **Q(12):** Handled the “no reservation” negation by using the set difference operator ($\setminus$). Filtered reservations with intervals overlapping November 6, 2026, projected `madeByPid`, and subtracted this set from the complete set of `pids` in `Person`.